

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H1



**ST. ANDREW'S JUNIOR COLLEGE
2014 JC2 Preliminary Examinations**

H1 BIOLOGY

8875/1

Paper 1: Multiple Choice

Thursday

18 September 2014

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser (not supplied)
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the multiple choice answer sheet in the spaces provided.

There are **30** questions in this paper. Answer all questions. For each question, there are four possible answers, A, B, C and D.

Choose the one you consider correct and record your choice in soft pencil on the separate multiple choice answer sheet.

INFORMATION TO CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for wrong answer. Any rough working should be done in this booklet.

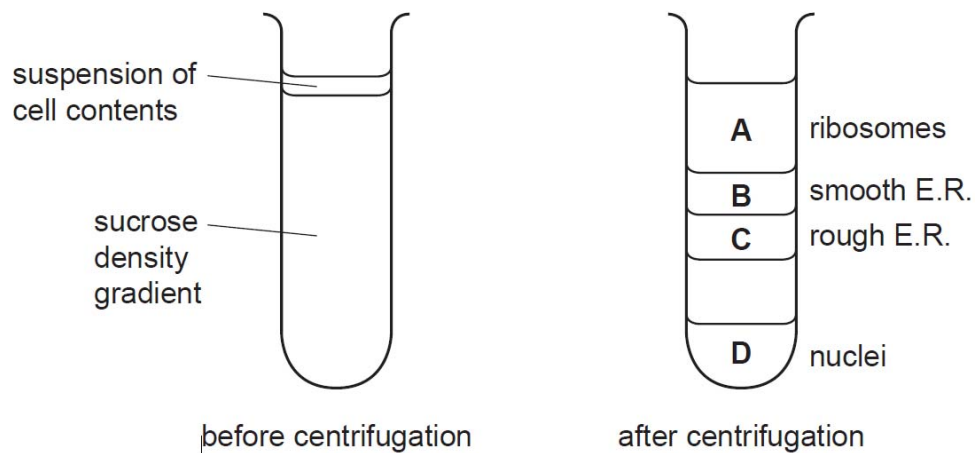
At the end of the examination, submit both question paper and multiple choice answer sheet.

This document consists of **21** printed pages.

[Turn over

- 1 Sometimes, scientists need to isolate organelles. This can be achieved by taking a number of cells and breaking their cell surface membranes to release the contents of the cells into a buffer solution.

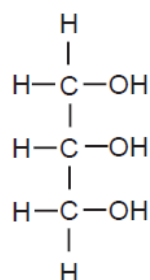
In zonal centrifugation, the suspension of cell contents is placed on top of a sucrose density gradient. The tube is then placed in a centrifuge and spun at high speed. The larger and denser particles will move towards the bottom of the tube faster than smaller and less dense particles as shown below.



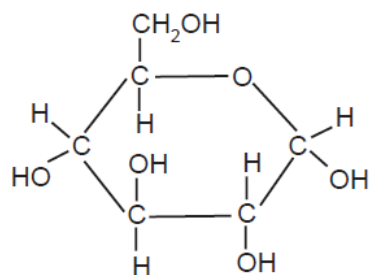
If a sample of intact prokaryotes had been added to a suspension of eukaryotic cell contents, where would you expect them to be found? **D**

2 Which molecule is found in glycogen? **B**

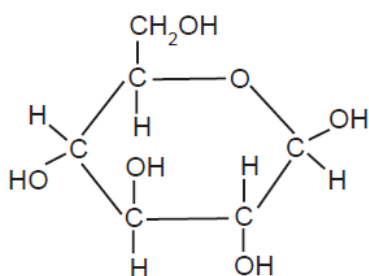
A



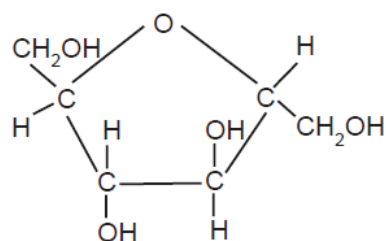
B



C

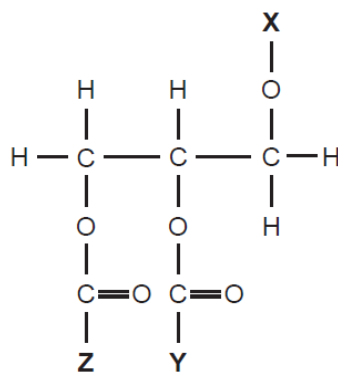


D



3 The diagram below shows a phospholipid molecule.

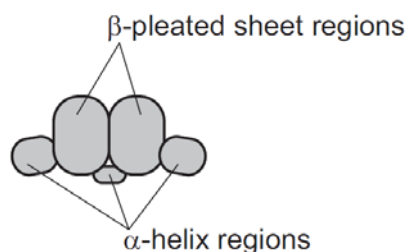
What are X, Y and Z?



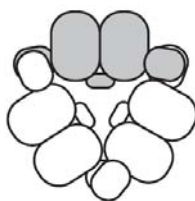
	X	Y	Z
A	Hydrocarbon chain	Hydrocarbon chain	Phosphate-containing group
B	Hydrocarbon chain	Glycerol	Phosphate-containing group
C	Phosphate-containing group	Hydrocarbon chain	Hydrocarbon chain
D	Phosphate-containing group	Glycerol	Hydrocarbon chain

4 Approximately half of the total protein in a pea seed consists of the storage protein vicilin.

- Each molecule of vicilin is made up of three identical polypeptides.
- Each polypeptide is made up of two β -pleated sheet regions with linking α -helix regions, folded into the shape as shown below.



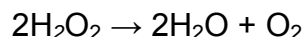
- This allows the three polypeptides to pack together into a compact, flat storage molecule, as shown below.



Which row correctly describes the structures of vicilin?

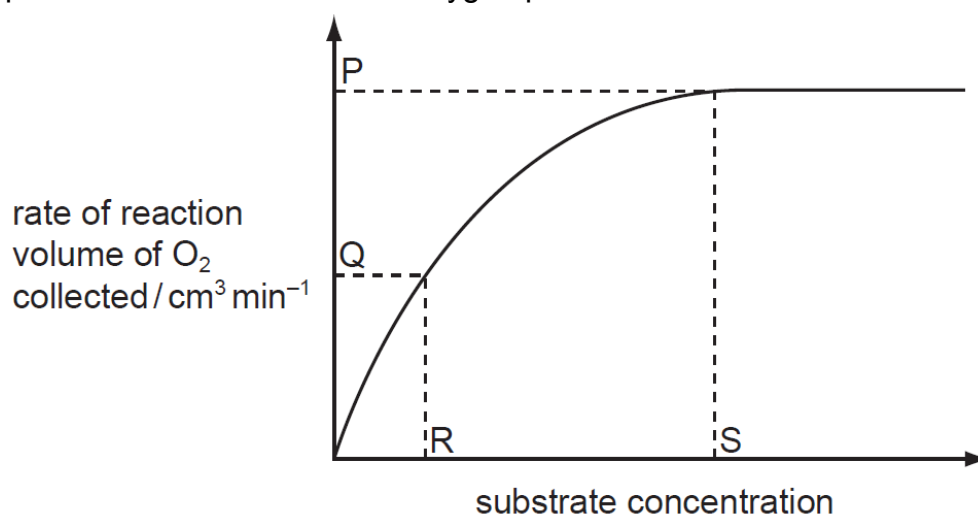
	Primary structure	Secondary structure	Tertiary structure	Quaternary structure
A	Amino acid sequence of one polypeptide	α -helix and β -pleated sheet regions of each polypeptide	Association of three polypeptides	Folding of each polypeptide
B	Amino acid sequence of one polypeptide	α -helix and β -pleated sheet regions of each polypeptide	Folding of each polypeptide	Association of three polypeptides
C	Association of three polypeptides	Amino acid sequence of one polypeptide	α -helix and β -pleated sheet regions of each polypeptide	Folding of each polypeptide
D	Association of three polypeptides	Amino acid sequence of one polypeptide	Folding of each polypeptide	α -helix and β -pleated sheet regions of each polypeptide

- 5 Liver tissue produces an enzyme called catalase which breaks down hydrogen peroxide into water and oxygen.



The rate of this reaction can be determined by measuring the volume of oxygen produced in a given length of time.

Students added small cubes of fresh liver tissue to a range of hydrogen peroxide solutions and measured the volumes of oxygen produced. Their data were used to produce the graph showing how changing the concentration of hydrogen peroxide affected the rate of oxygen production.



Which statements are correct?

- 1 At P, the rate of reaction is limited by the concentration of enzyme.
- 2 At Q, all of the enzyme active sites are occupied by substrate molecules.
- 3 At Q, the rate of reaction is limited by the concentration of the substrate.
- 4 R represents K_m where the reaction rate = $V_{\text{max}} / 2$
- 5 At S, all of the enzyme active sites are occupied by substrate molecules.

- A** 1, 3, 4 and 5
B 1, 4 and 5
C 2 and 3
D 2 and 5

- 6 A short piece of DNA 19 base pairs long was analysed to find the number of nucleotide bases in each of the polynucleotide strands. Some of the results are shown below.

	number of nucleotide bases			
	A	C	G	T
strand 1				4
strand 2		7		5

How many nucleotides containing C were present in strand 1?

- A 2
B 3
 C 5
 D 7

- 7 When DNA replicates, new nucleotides containing the common isotope of nitrogen (^{14}N) are used to build new nucleic acids.

In the laboratory, nucleotides can be synthesised using the heavy isotope of nitrogen (^{15}N). DNA is allowed to replicate once using these ^{15}N nucleotides, then twice more using ^{14}N nucleotides.

What will be the ratio of ^{14}N to ^{15}N nucleotides in the final molecules?

- A 1:1
 B 3:1
 C 5:1
D 7:1

8 Which statements correctly describe the structure and function of prokaryotic ribosomes?

- 1 Prokaryote ribosomes are smaller than eukaryote ribosomes and sediment at 70S.
- 2 A prokaryote ribosome consists of two subunits, one of 50S and one of 30S.
- 3 In prokaryotes, ribosomes translate mRNA in the same cellular compartment in which it is transcribed.
- 4 In prokaryotes, ribosomes can begin translating mRNA before its synthesis has been completed.
- 5 A prokaryote ribosome can accommodate only one aminoacyl-tRNA at a time.

- A 1, 2, 3, 4 and 5
B 1, 2, 3 and 4
C 1, 2 and 4
D 1, 3 and 4

- 9 Ribonuclease is an enzyme that digests RNA. The first five amino acids of the functioning molecule of ribonuclease are :

lys – glu – thr – ala – ala

The mRNA of the gene coding for ribonuclease, for the first 15 nucleotides, has the following sequence,

AUGAAGGAAACUGCU

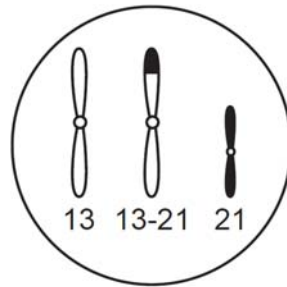
A genetic code, showing mRNA codons, is shown below.

first position	second position				third position
	U	C	A	G	
U	phe	ser	tyr	cys	U
	phe	ser	tyr	cys	C
	leu	ser	STOP	STOP	A
	leu	ser	STOP	trp	G
C	leu	pro	his	arg	U
	leu	pro	his	arg	C
	leu	pro	gln	arg	A
	leu	pro	gln	arg	G
A	ile	thr	asn	ser	U
	ile	thr	asn	ser	C
	ile	thr	lys	arg	A
	met	thr	lys	arg	G
G	val	ala	asp	gly	U
	val	ala	asp	gly	C
	val	ala	glu	gly	A
	val	ala	glu	gly	G

Which event occurs to explain the information given above?

- A** The first amino acid on the polypeptide chain is removed in post-translational modification.
- B** The first codon is removed from the mRNA transcript in post-transcriptional modification.
- C** The mRNA binds to the rRNA in the second codon position.
- D** There is no tRNA with an anticodon complementary to the first codon.

- 10 Down's syndrome can be caused by a trisomy of chromosome 21, but can also result from translocation of chromosome 21 onto chromosome 13, forming a single chromosome 13-21.



The diagram shows chromosomes 13 and 21 in the nucleus of a diploid ($2n$) testis cell from a phenotypically normal male carrier of a 13-21 translocation. This cell has a chromosome number of 45.

Which is not a likely outcome of fertilisation of normal oocytes by sperm from this male?

	Chromosomes in sperm	Embryo
A	13 and 21	$2n = 46$ normal phenotype
B	13-21	$2n = 45$ normal phenotype
C	13-21 and 21	$2n = 46$ Down's syndrome
D	13-21 and 21	$2n = 47$ Down's syndrome

- 11 In fruit flies, one gene controls wing form (normal or vestigial) and one gene controls eye colour (red or normal brown).

A fly with normal wings and normal brown eyes is crossed with a fly with vestigial wings and red eyes. All the F_1 are normal for both characteristics.

However, when F_1 are crossed with each other, the resulting F_2 is :

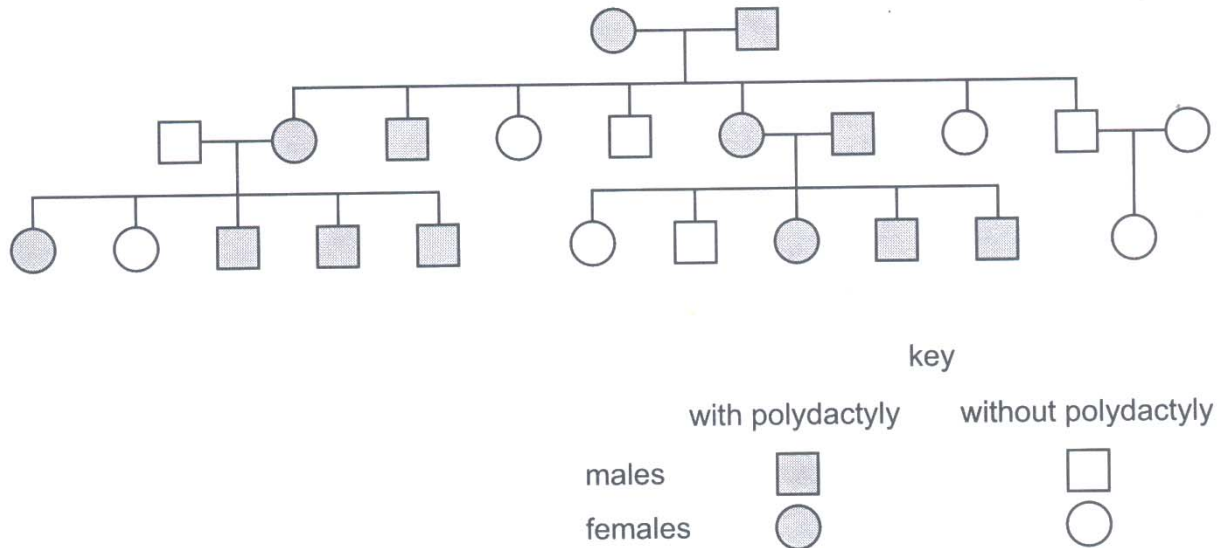
45 normal wing, normal brown eye
 17 normal wing, red eye
 16 vestigial wing, normal brown eye
 5 vestigial wing, red eye
 1 normal wing, orange eye

What is the best explanation for the results of this dihybrid cross?

- A** Codominance
B Gene mutation
C Multiple alleles
D Sex linkage

- 12** Polydactyly is a genetic condition, controlled by a single gene, in which people are born with extra fingers and toes.

The diagram shows the pedigree of a family.



What is true about the inheritance of polydactyly?

	The allele coding for polydactyly is dominant	The allele coding for polydactyly is autosomal.	Offspring of parents with polydactyly will all inherit the allele.	More males than females will inherit the polydactyly allele
A	True	True	False	False
B	True	False	False	True
C	False	False	False	True
D	False	False	True	True

13 Two genes involved in coat colour of goats are at loci on different chromosomes.

The colour gene C causes the hairs to have uniform colour and has three alleles.

- C^{DB} giving dark brown hairs
- C^B giving black hairs
- C^{MB} giving medium brown hairs

A dominant allele of the agouti gene (A^G) causes the development of white hairs between the colour hairs giving the coat a shaded appearance

The table shows the results of crosses between a male goat and two female goats.

Parents	Offspring
Black agouti male X Uniformly dark brown female	50% uniform, 50% agouti ; 50% dark brown, 50% black
Black agouti male X Medium brown agouti female	All agouti ; 50% black, 50% medium brown

Which row shows the possible genotypes of the male and female goats?

	Black agouti male	Uniformly dark brown female	Medium brown agouti female
A	$C^B C^{MB} A^G A^G$	$C^B C^{DB} A^G A^g$	$C^{MB} C^{MB} A^G A^G$
B	$C^B C^{MB} A^G A^g$	$C^B C^{DB} A^g A^g$	$C^B C^{MB} A^G A^G$
C	$C^B C^{DB} A^G A^G$	$C^{DB} C^{MB} A^G A^g$	$C^B C^{DB} A^G A^g$
D	$C^B C^{MB} A^G A^g$	$C^B C^{DB} A^g A^g$	$C^{MB} C^{MB} A^G A^G$

- 14** Albino horses have recessive alleles which cannot be transcribed to produce pigmentation. Geneticists studying horses have found a different allele of the same gene which predisposes carriers to premature ageing of the hair follicles. This allele is dominant, so, even when born with pigmented hair, the coats of horses with this allele will gradually turn white over the first 6 to 8 years of life.

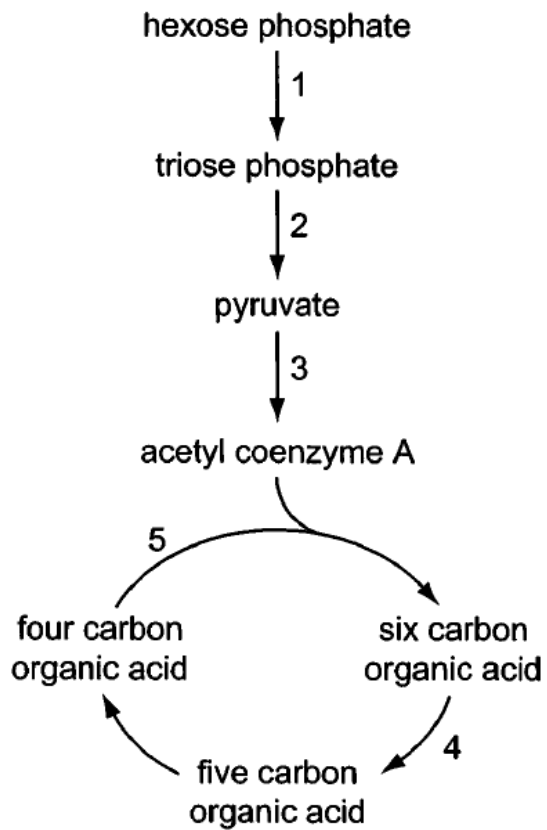
Which **white** female horse would be best to use in crosses to determine the genotype of a ten-year-old, white male horse?

- A** An unrelated, ten-year-old female
- B** An unrelated, three-year-old female
- C** A sister of the male
- D** The male's mother

- 15** What are the features of photophosphorylation (PP) and oxidative phosphorylation (OP)?

	ATP produced	Electrons flow	Oxygen is final electron acceptor	Can be cyclic
A	PP and OP	PP and OP	OP only	PP only
B	PP and OP	OP only	PP and OP	OP only
C	PP only	PP only	OP only	PP and OP
D	OP only	PP and OP	PP only	PP only

16 The flow diagram shows some of the stages in respiration.



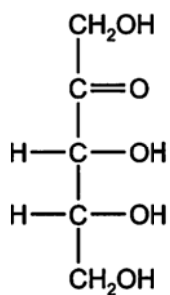
During which stages does oxidative decarboxylation occur?

- A 1 and 3
- B 2 and 3
- C 3 and 4**
- D 4 and 5

17 Which molecule is a co-factor used by dehydrogenase enzymes?

- A Coenzyme A
- B Cytochrome
- C Lactate
- D NAD**

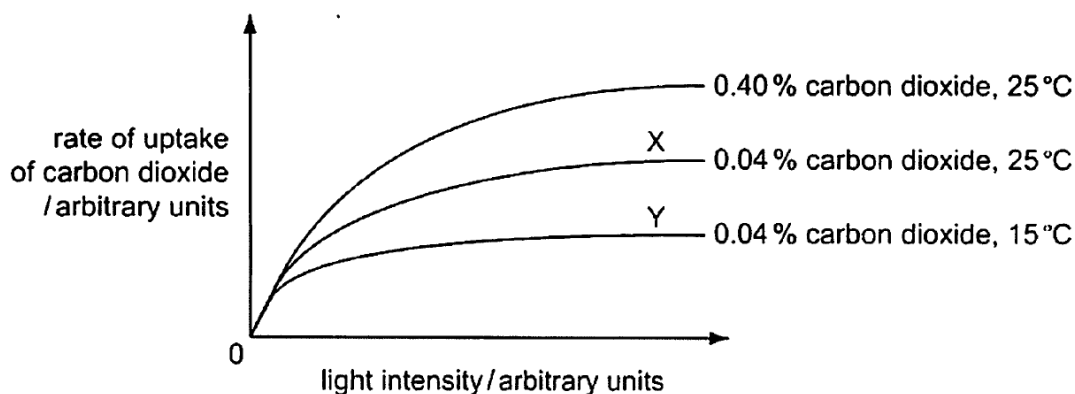
18 More than 80% of the first formed sugar in photosynthesis is converted into the phosphorylated form of the compound shown.



What is the compound?

- A Adenosine
- B Deoxyribose
- C Ribose
- D Ribulose**

- 19 The graph shows the rate of uptake of carbon dioxide by a photosynthetic plant in different conditions.



Which processes are shown by these curves to limit the rate of uptake of carbon dioxide?

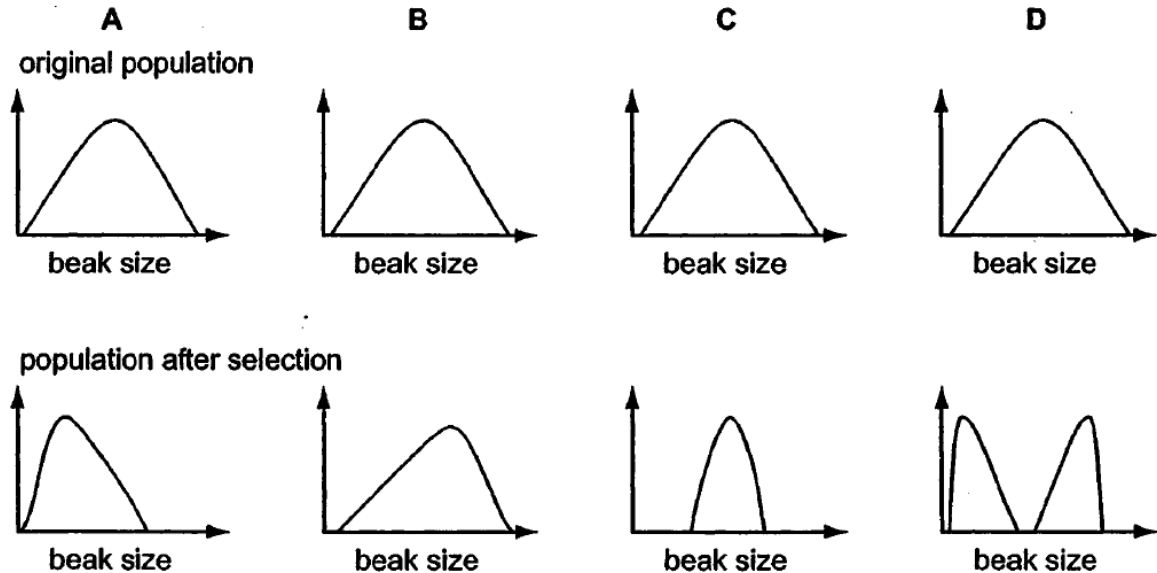
	X	Y
A	Light dependent reaction	Light dependent reaction
B	Light dependent reaction	Light independent reaction
C	Light independent reaction	Light dependent reaction
D	Light independent reaction	Light independent reaction

- 20 Which statement about the fluid mosaic model of membrane structure is correct?

- A** The less unsaturated the fatty acids tail of the phospholipid, the more fluid the membrane.
- B** The more unsaturated the fatty acids tail of the phospholipid, the more fluid the membrane.
- C** The higher the temperature, the less fluid the membrane.
- D** The lower the temperature, the more fluid the membrane.

- 21 One species of finch on an island shows variation in beak size. Birds with larger beaks can eat larger seeds. Following a dry period, fewer small seeds are available, with larger seeds and nuts being more plentiful.

Which graph shows the effect of natural selection on beak size of these birds following a dry period? **B**



- 22 Which statements are acceptable parts of Darwinian evolutionary theory?

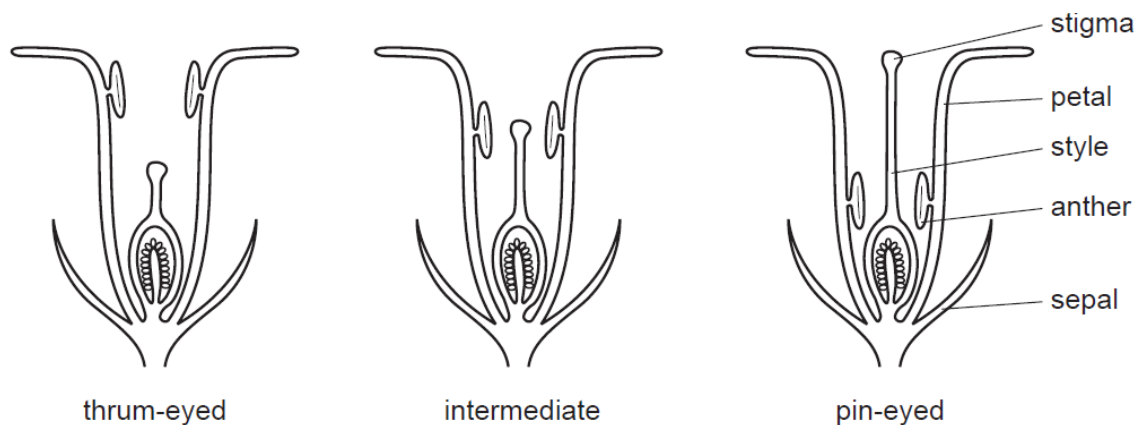
- 1 Advantageous behaviour acquired during the lifetime of an individual is likely to be inherited.
- 2 In competition for survival, the more aggressive animals are more likely to survive.
- 3 Species perfectly adapted to a stable environment will continue to evolve.
- 4 Variation between individuals of a species is essential for evolutionary change.

- A** 1, 2 and 4
B 2 and 3
C 3 and 4
D 4 only

- 23** The primrose, *Primula vulgaris*, is a small herbaceous, yellow-flowered plant which is common in cooler areas of the Northern hemisphere including alpine and Arctic areas.

The flowers of the primrose have different flower shapes (polymorphic), which are adaptations for pollination. 'Thrum-eyed' primroses have a short style. 'Pin-eyed' primroses have much longer styles. Some populations of primrose consist almost entirely of plants with intermediate flowers. These populations are common where there are fewer winged insects.

The diagrams show polymorphic flowers of primroses.

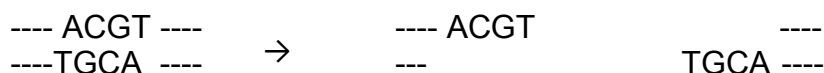


- 1 Cross-pollination will be favoured between pin-eyed and thrum-eyed primroses.
- 2 Primroses with pin-eyed flowers are likely to show more genetic diversity than primroses with intermediate flowers.
- 3 Primroses with intermediate flowers are likely to be more able to adapt to changing environmental conditions than pin-eyed and thrum-eyed primroses.
- 4 Self-pollination is more likely to occur in primroses with intermediate flowers.

Which statements are correct?

- A 1 and 2
- B 1, 2, 3 and 4
- C 1, 2 and 4**
- D 3 and 4

- 24 The restriction enzyme *Tai*1, cuts DNA as shown.



Which of the plasmids below can be cut using restriction enzyme *Tai*1, in order to allow DNA to be inserted into the circular plasmid without the loss of any part of the plasmid?

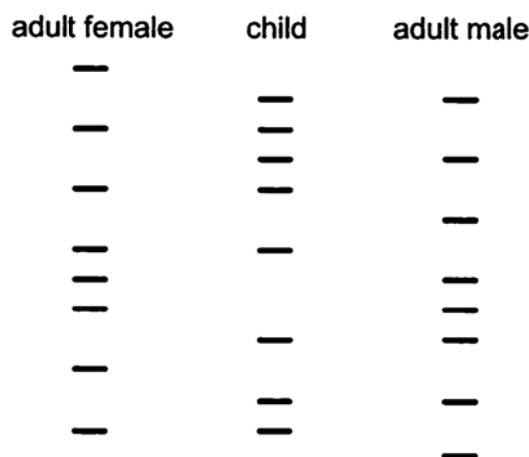
- A** ---- TAACGTAC ---- CTCAAGCT ----
 ---- AATGCATG ---- GAGTTCGA ----
- B** ---- TTAACGTA ---- CACGTGGT ----
 ---- AATTGCAT ---- GTGCACCA ----
- C** ---- ACGTACGT ---- TCCACGTA ----
 ---- TGCATGCA ---- AGGTGCAT ----
- D** ---- ACCTAGGT ---- CCACCTGA ----
 ---- TGGATCCA ---- GGTGGACT ----

Key

---- other nucleotides

The rest of the plasmid does not contain this restriction site.

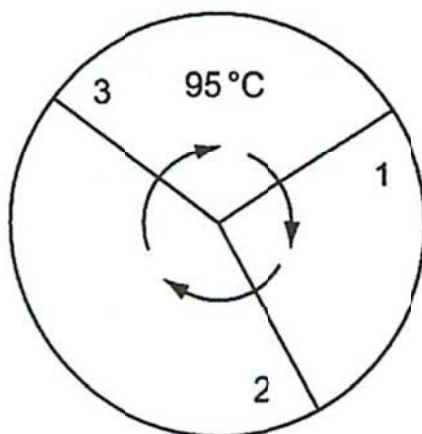
- 25 The diagram shows the results of a DNA analysis, carried out by electrophoresis. The DNA of three individuals was profiled.



What is the most likely conclusion from these results?

- A** The child cannot be related to either the male or female.
B The female cannot be the child's mother.
C The male cannot be the child's father.
D The male and female could be the child's parents.

26 The diagram represents the stages of a polymerase chain reaction.



Which row shows the events happening at stages 1, 2 and 3?

	1	2	3
A	Binding of primers at 55°C	Polymerisation at 72°C	Denaturing of DNA
B	Denaturing of DNA at 72°C	Binding of primers at 55°C	Polymerisation
C	Polymerisation at 55°C	Binding of primers at 72°C	Denaturing of DNA
D	Polymerisation at 55°C	Denaturing of DNA at 72°C	Binding of primers

27 Which uses of the information from the human genome project are generally considered to be unethical?

- 1 An insurance company only giving cheap rates to people with genetic predispositions to fewer diseases
- 2 Genetic archaeologists identifying the earliest forms of genes to show evolutionary relationships
- 3 Cytologists developing test for only some defective genes
- 4 Doctors only giving specific drugs to block the actions of faulty genes to carriers of those genes
- 5 Genetic counsellors giving specific lifestyle information only to people genetically predisposed to risks
- 6 Parents choosing embryos for implantation only after ante-natal tests for acceptable genes

- A** 1 and 3
B 1 and 6
C 2 and 5
D 3 and 4

- 28** Stem cells are found in many tissues that require frequent cell replacement such as the skin, the intestine or the blood.

However, within their own environments, a bone marrow cell cannot be induced to produce a skin cell and a skin cell cannot be induced to produce a bone marrow cell.

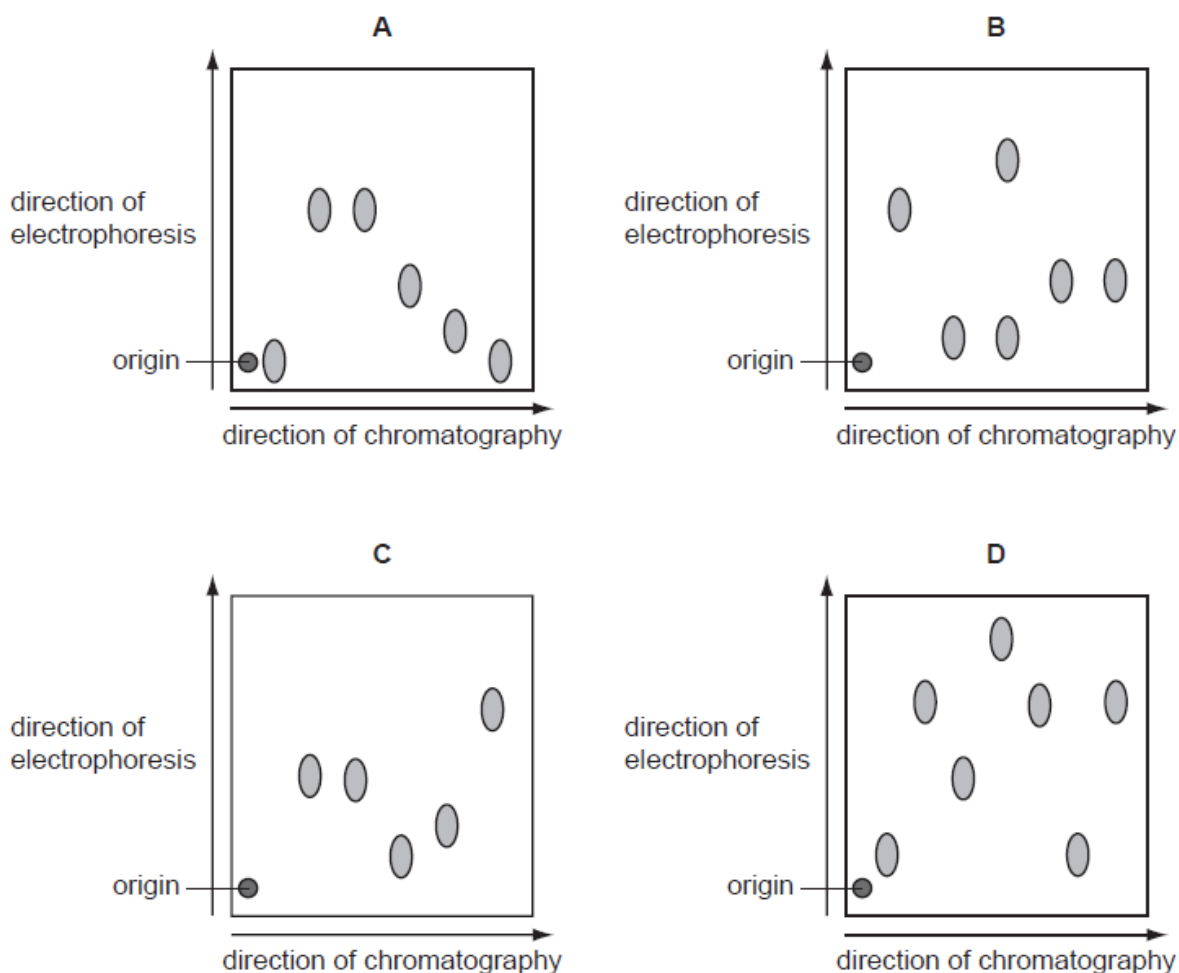
Which statement explains this?

- A** Different stem cells have only the genes required for their particular cell line.
 - B** Genes not required for a particular cell line are methylated.
 - C** Genes not required for a particular cell line are removed.
 - D** mRNA that is not required for a particular cell line is destroyed.
- 29** Which genetic modification would enable higher yield in crops grown on fertile soil in a tropical region?

- A** Drought tolerance
- B** Increased carbon fixation
- C** Increased vitamin A content
- D** Salt tolerance

- 30 The diagrams show the results of an investigation into the composition of different mixtures of amino acids. Each mixture of amino acids was separated using chromatography. Each chromatogram was then turned through 90° and the amino acids separated again by electrophoresis.

Which diagram shows an amino acid mixture in which the solubility of some of the amino acids is the same but the charge on those particular amino acids is different? **B**



- END OF PAPER -